

Problem 45. Let a, b, c be distinct numbers such that the equations $x^2 + ax + 1 = 0$ and $x^2 + bx + c = 0$ have a common real root, and the equations $x^2 + x + a = 0$ and $x^2 + cx + b = 0$ also have a common real root. Find the value of $a + b + c$.

Solution. Let t be the common real root of $x^2 + ax + 1 = 0$ and $x^2 + bx + c = 0$. Then we have

$$t^2 + at + 1 = 0, \quad (1)$$

$$t^2 + bt + c = 0.$$

Taking their difference, we get $(a - b)t = c - 1$. As a and b are distinct, this gives $t = \frac{c-1}{a-b}$. Similarly, let u be the common real root of $x^2 + x + a = 0$ and $x^2 + cx + b = 0$. Then we have

$$u^2 + u + a = 0, \quad (2)$$

$$u^2 + cu + b = 0.$$

Taking their difference, we get $(c - 1)u = a - b$. If $c = 1$, then we have $a = b$, which is impossible. So $u = \frac{a-b}{c-1} = \frac{1}{t}$. Since the product of roots of equation (1) is 1, the roots of this equation are t and $\frac{1}{t}$. Hence, we have

$$t + \frac{1}{t} = -a. \quad (3)$$

Similarly, by considering equation (2), its roots are $\frac{1}{t}$ and at , and hence we have

$$\frac{1}{t} + at = -1. \quad (4)$$

Taking the difference of (3) and (4), we get $(a - 1)t = a - 1$. Note that $a \neq 1$ since otherwise (1) has no real root. So $t = 1$. Putting $t = 1$ in (1), we get $a = -2$. Putting $t = 1$ in $t^2 + bt + c = 0$, we get $b + c = -1$. So $a + b + c = (-2) + (-1) = -3$. $\square$

Problem 46. Einstein chose a natural number $X \leq 100$, and Pythagoras is trying to guess this number. He can select two natural numbers M and N less than 100 and ask about $\gcd(X + M, N)$. Show that Pythagoras can determine Einstein's number with at most seven questions.

Solution 1. For the first question, Pythagoras can select $M = 1$ and $N = 2$. If $\gcd(X + 1, 2) = 1$, then X is even. If $\gcd(X + 1, 2) = 2$, then X is odd. Suppose Pythagoras finds out $X \equiv r \pmod{2^k}$ after asking the k th question, where $k \leq 5$ and $0 \leq r < 2^k$. For the $(k + 1)$ st question, Pythagoras can select $M = 2^k - r$ and $N = 2^{k+1}$. Since $X + M \equiv r + (-r) = 0 \pmod{2^k}$, the gcd can only be 2^k or 2^{k+1} . If it is 2^k , then $X \equiv r \pmod{2^{k+1}}$. If it is 2^{k+1} , then $X \equiv 2^k + r \pmod{2^{k+1}}$. Therefore, Pythagoras can find out the remainder r when X is divided by 64 after asking 6 questions.

If $r = 0$ or $37 \leq r \leq 63$, then X is known since $X \leq 100$. If $1 \leq r \leq 36$, then X can be r or $r + 64$. Suppose $r \equiv k \pmod{3}$ where $k = 0, 1, 2$. Then Pythagoras can select $M = 3 - k$ and $N = 3$ in the 7th question. If the gcd is 1, then $X = r + 64$. If it is 3, then $X = r$.

In total, at most seven questions are needed. $\square$

Solution 2. For the first four questions, Pythagoras can select $M = 1, 2, 3, 4$ respectively and $N = 60$. Then $X \equiv r \pmod{3}$ where $r = 0, 1, 2$ if and only if $\gcd(X + (3 - r), 60)$ is divisible by 3. Similarly, $X \equiv r \pmod{4}$ where $r = 0, 1, 2, 3$ if and only if $\gcd(X + (4 - r), 60)$ is divisible by 4. Also, $X \equiv r \pmod{5}$ where $r = 1, 2, 3, 4$ if and only if $\gcd(X + (5 - r), 60)$ is divisible by 5. In case none of these gcd's is divisible by 5, then we must have $X \equiv 0 \pmod{5}$. Therefore, after asking four questions, Pythagoras can find out the remainder when X is divided by each of 3, 4 and 5.

Suppose $X \equiv r \pmod{4}$ where $r = 0, 1, 2, 3$. For the fifth question, Pythagoras can select $M = 4 - r$ and $N = 8$. If $\gcd(X + (4 - r), 8)$ equals 4, then $X \equiv r \pmod{8}$. If it equals 8, then $X \equiv r + 4 \pmod{8}$. Now, Pythagoras can find out the remainder when X is divided by 8. By the Chinese remainder theorem, Pythagoras can determine $X \pmod{120}$. As $X \leq 100$, the value of X is known.

In total, at most five questions are needed. $\square$

$$\angle CH'A = \angle CBA = 180^\circ - \angle B'BC = 180^\circ - \angle B'C'C = \angle AC'D,$$

Problem 48. Let A, B, C, a, b and c be positive integers. Suppose that the sum $S = A + B + C + a + b + c$ divides both $ABC + abc$ and $AB + BC + CA - ab - bc - ca$. Prove that S is composite.

$$P(x) = Sx^2 + (AB + BC + CA - ab - bc - ca)x + (ABC + abc).$$

From the given conditions, all coefficients of $P(x)$ are divisible by S . Therefore, $S \mid P(x)$ for all integers x . In particular, by taking $x = c$, we have $S \mid P(c) = (c + A)(c + B)(c + C)$. If S is a prime, then S divides one of $c + A$, $c + B$ or $c + C$. But this is impossible since S is greater than all of them and all of them are positive. Therefore, S must be composite. $\square$